

Research Paper

Genome-wide identification of the pigment formation-regulating *CYP450* family gives new insights into color improvement in *Bougainvillea*

Rong Sun ^{*}, Shan Liu, Yi Zheng

Department of Biological Engineering, College of Biological and Chemical Engineering, Panzhihua University, Panzhihua 617000, PR China

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ABSTRACT

Bougainvillea is a popular ornamental plant, with modification of flower color is a primary breeding objective. Flower color can be attributed to three classes of pigment: flavonoids, carotenoids, and betalains. The cytochrome P450 monooxygenase gene family plays important roles in pigment biosynthesis. In *Bougainvillea*, although the CYPs involved in betalains biosynthesis has been reported, the CYPs related to flavonoids and carotenoids biosynthesis have not been characterized. Here, whole-genome analyzes identified 270 *BbCYPs* in *Bougainvillea*, which were divided into eight clans via phylogenetic analysis. By exploring the structure, physicochemical properties, chromosome localization and motif composition of *BbCYPs*, as well as evolutionary relationships with CYPs in other plants, we preliminarily screened 13 potential *BbCYPs* related to pigment biosynthesis. The expression levels of these genes except *BbCYP1*, at different developmental stages showed an upward trend. Correlation analysis results indicated that betalains and flavonoids may play major roles in *Bougainvillea* × *buttiana* 'Mrs Butt' bracts color formation. Among 13 *BbCYPs*, the *BbCYP40* and *BbCYP220* genes exhibited the strongest correlation with betalains and flavonoids, respectively. *BbCYP85* gene exhibited stronger correlation with carotenoids than *BbCYP162* gene. These findings provide valuable genetic resources for improving *Bougainvillea* bract color.

1. Introduction

Bougainvillea, native to South America, is a renowned ornamental plant widely cultivated in tropical and subtropical regions, including China (Wu and Tang, 2010). The color of its bracts is a primary ornamental trait, with various cultivars classified based on bract type, including monochrome single, monochrome double, double colored single, and single colored spotted bract (Sindhu et al., 2020). Among them, monochrome single bract cultivars are the most prevalent, featuring purple, red, pink, orange, yellow, and white bracts (Zeng et al., 2018). Considering its ornamental and commercial importance in nursery trade, developing novel colors as a crucial breeding objective for this ornamental plant (Chen et al., 2009; Du et al., 2024; Kang et al., 2022).

The color of plant flowers is primarily determined by three pigment classes: flavonoids, carotenoids, and betalains. A subclass of flavonoids, the anthocyanins, produce a wide range of colors from orange to red to violet and blue. Currently, the most well-studied anthocyanins include petunidin, pelargonidin, delphinidin, cyanidin, peonidin, and malvidin,

with delphinidin being associated with blue flower formation (Tanaka and Brugliera, 2013). Carotenoids, which include carotenes and lutein, mainly yield yellow or red flower colors, such as in sunflowers, chrysanthemums, and roses (Tanaka et al., 2008). Betalains consist of betacyanin and betaxanthin, with betacyanin conferring red, purple, pink, and betaxanthin conferring yellow and orange flower colors (Tanaka et al., 2008). These pigments are synthesized via complex, multi-branched pathways involving several key enzymes, with the cytochrome P450 monooxygenase (CYP) superfamily playing a crucial role (Tanaka, 2006).

Cytochrome P450 enzymes (P450 s) are known to catalyze various reactions in plant biosynthetic pathways and play a crucial role in the biosynthesis of secondary metabolites, including anthocyanin, carotenoids, and betalains (Hallahan and West, 1995; Li et al., 2012; Schoenbohm et al., 2000). Based on their roles, P450 s can be classified into four categories, each containing multiple families (Mizutani, 2012). In category IV, the CYP75 family members flavonoid 3'-hydroxylase (F3'H) and flavonoid 3',5'-hydroxylase (F3'5'H) determine the hydroxylation pattern of the B-ring of the anthocyanin molecule and, thus, flower color

* Corresponding author.

E-mail address: sunrong@pzu.edu.cn (R. Sun).

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(Yoshida et al., 2009). The CYP76 family member CYP76AD is involved in betacyanin biosynthesis, catalyzing a biosynthetic step from L-DOPA to cycloDOPA (Hatlestad et al., 2012). In category I, the CYP97 family member CYP97A3 primarily catalyzes β -hydroxylation of α -carotene in *Arabidopsis* (Kim and DellaPenna, 2006).

Recent advancements in molecular biology and plant transformation technology have enabled the modification of flower color through genetic engineering, with modulation of P450 gene expression being a common strategy. For example, overexpression of *F3H* gene in *torenia* resulted in darker pink flowers due to increased cyanidin production (Ueyama et al., 2002). Expression of petunia *F3'5'H* cDNAs in a pale pink petunia deficient in *F3'5'H* and *F3H* activities resulted in reddish-purple transgenic plants (Holton et al., 1993). Heterologous expression of the *viola F3'5'H* gene in rose cultivars resulted in a novel bluish flower color with elevated delphinidin accumulation (up to 95 %) (Katsumoto et al., 2007). Overall, flower color modification via P450 sequences has been previously used in many plants to obtain numerous novel-colored products.

To cultivate novel-colored *Bougainvillea*, although we identified some key pigment synthesis-related enzyme-coding genes, CYP family remained unidentified. In this study, 270 *BbCYP* genes were identified at the genome-wide level, including 13 *BbCYP* genes that potentially regulate pigment synthesis based on phylogenetic analysis, gene structure analysis, motif analysis, and evolutionary relationship with other species. Based on the association analysis among gene expression levels, pigments contents, and changes in color parameters at different developmental stages, three candidate genes were identified as being closely associated with the pigment synthesis. Therefore, our findings provide valuable information for breeding novel-colored *Bougainvillea*.

2. Material and methods

2.1. Plant materials

Bougainvillea cultivar *Bougainvillea* × *buttiana* ‘Mrs Butt’ (BBMB) bracts were collected from the nursery of Panzhihua University at different developmental stages (S1: tender, S2: beginning flowering, and S3: blooming) for gene expression analysis, pigments contents, and color parameter determination.

2.2. Characterization bougainvillea CYP genes

The genome assembly sequences and gene annotations data of BBMB were obtained from the China National GeneBank (<https://www.cnbg.org/>) (Lan et al., 2023). The hidden Markov model (HMM) profile of the P450 domain was retrieved from the Pfam protein family database by ID (PF00067) (<http://pfam.sanger.ac.uk/>). Using the Simple HMM Search tool of TBtools software, inputting the BBMB protein sequences, Pfam model, and Pfam ID (PF00067) to search candidate BBMB CYP450 genes by domain alignment (Chen et al., 2023). Candidate CYPs were screened based on E-value ($<1e-40$) of sequence score and structural domain score. BLASTp queries were then performed on the NCBI website to confirm the membership of candidate CYPs in the CYP gene family. Finally, their amino acid number, molecular weight (MW), isoelectric point (pI), instability index, aliphatic index and grand average of hydropathicity were calculated using the TBtools Protein Paramter Calc program.

2.3. Phylogenetic analysis

To classify and filter candidate CYPs, a neighbor-joining (NJ) systematic evolutionary tree was constructed using CYP proteins from *Bougainvillea* × *buttiana*, *Arabidopsis thaliana*, *Glycyrrhiza echinata*, *Beta vulgaris*, *Glycine soja*, and *Dianthus caryophyllus*. ClustalX 1.81 was used for sequence alignments of CYP proteins in these plant species. The systematic evolutionary tree was built using MEGA 5.05 software with

the NJ method and 1000 bootstrap replications (Tamura et al., 2011). The phylogenetic tree of *BbCYP*s was constructed using the same methodology.

2.4. Gene structure, motif composition and conserved domains analysis

The gene structure of *BbCYP* genes was visualized using TBtools software, followed by determination of conserved motifs in *BbCYP* proteins using the MEME online program (<https://meme-suite.org/meme/tools/meme>) with a set parameter of 10 motifs in classic mode. Simultaneously, the conserved domains of *BbCYP* proteins were identified using the Batch CD-Search tool. (<https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>). Finally, the Gene Structure View (Advanced) program in TBtools was used to generate a single image displaying the phylogenetic tree, gene structure, motifs, and domains of *BbCYP*s. The results provided a comprehensive overview of the structural features of *BbCYP* genes and proteins.

2.5. Chromosomal mapping and syntenic analysis with other plant species

The physical location information obtained from the BBMB genomic annotation library was used to map all the *BbCYP* genes on the chromosomes using the TBtools "Gene Location Visualize from GIF/GFF" feature. The Multiple Collinear Scanning Toolkits (MCScanX) were employed to analyze the tandem and segmental replication events in the *BbCYP* genes with default parameters (Wang et al., 2012). The collinear relationships between CYP genes from BBMB and other plants (*G. max*, *A. thaliana*, *B. vulgaris*, *N. tabacum*, and *O. sativa*) were revealed using the TBtools "Dual System Plot" feature.

2.6. Quantitative RT-PCR gene expression analysis

The expression profiles of various *BbCYP* genes in bracts during three developmental stages were analyzed using quantitative real-time PCR (qRT-PCR). The RNAprep pure Plant Kit and the FastKing gDNA Dispelling RT SuperMix Kit purchased from TIANGEN BIOTECH (BEIJING) CO., LTD were used to extract total RNA and reverse transcribe single-stranded cDNA, respectively. The internal reference gene was F1-ATPase alpha subunit (ATP1) (Sun et al., 2024). The qRT-PCR primers for *BbCYP* genes were designed based on sequences obtained from the BBMB genomic database (Table S1). The lowest expression level among all samples of each gene was set to “1”. The qRT-PCR experiments were conducted according to the method described by Sun et al. (2024), with three biological replicates and three technical replicates.

2.7. Pigment contents measurement

The bracts of *Bougainvillea* at three developmental stages were used to determine flavonoids, carotenoids, and betalains contents. The extraction of pigments and determination of their content followed a previously described method (Zhong et al., 2022). Each experiment was replicated thrice.

2.8. Color parameter measurement

The Lightness (L), red-green phase value (a), yellow-blue phase value (b), Chroma (C*), and Hue (H*) of bracts at three developmental periods were measured using the COLOR MUSE SE color extractor (Variable Inc, Georgia) in a well-lit indoor environment without direct sunlight. Measurements were taken at the central part of the upper epidermal cells near the petiole at the lower end of the bract, and each bract was measured thrice (Zhong et al., 2022).

2.9. Statistical analysis

The qRT-PCR, pigment content, and color parameter measurement

experimental data were analyzed and plotted using Origin 2021 software for correlation analyzes and the least significant difference (LSD) test.

3. Results

3.1. Identification of BbCYP genes

A total of 313 CYP450 genes were identified from the genome of BBMB using TBtools "Simple HMM Search" function. After removing shorter and repeated sequences, 270 candidate CYPs were obtained and named BbCYP1 to BbCYP270 based on their chromosomal location. The encoded peptides varied in length from 375 to 630 amino acids, with BbCYP40 being the smallest protein (375 aa, 42.5 kDa) and BbCYP162 being the largest (630 aa, 70.6 kDa). The average pI of BbCYP proteins was 8.42, with BbCYP40 and BbCYP248 having the lowest and highest pI, respectively. 91.5 % of BbCYP proteins were hydrophilic. The detailed information is provided in Table S2.

3.2. Classification of BbCYP proteins

The CYP75A/B, CYP93B, CYP97A/C, and CYP76AD genes are implicated in pigment formation (Tanaka and Brugliera, 2014). To identify CYP genes related to pigment synthesis in *Bougainvillea*, we conducted a phylogenetic analysis comparing 270 BbCYPs with CYPs from *A. thaliana*, *G. echinata*, *B. vulgaris*, *G. soja* and *D. caryophyllus* (Fig. 1). The analysis classified these CYP genes into nine clans: CYP71, CYP51, CYP72, CYP74, CYP85, CYP86, CYP97, CYP710, and CYP711, with CYP71 being the sole A-type P450s. The CYP71 clan contained the highest number of proteins, while the CYP74 clan had the fewest. The distribution of BbCYP proteins across clans was as follows: 160 in CYP71, 39 in CYP72, 35 in CYP85, 26 CYP86, 4 in CYP51, 3 in CYP97, 2 in CYP711, and 1 in CYP710. Notably, no BbCYP proteins were found in the CYP74 clan. Among these, 13 BbCYP proteins are proposed to belong to the pigment synthesis-related clan (Fig. 1).

Similarly, from Fig. 2A, we can see that the BbCYP proteins also are organized into only 8 clans. And the CYP71, CYP85, CYP72, and CYP86 clans can be further divided into many subfamilies. The CYP71 clan

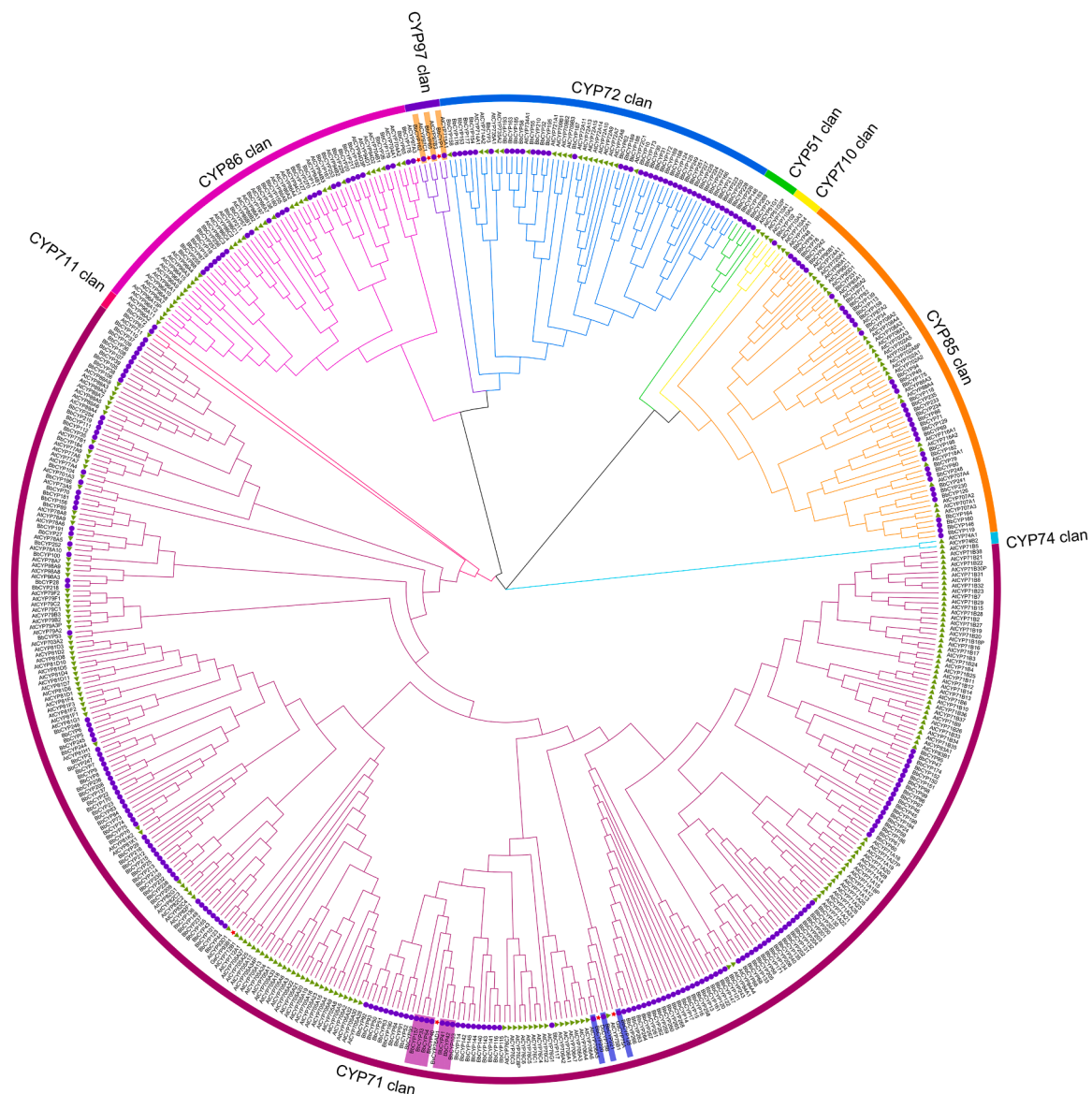
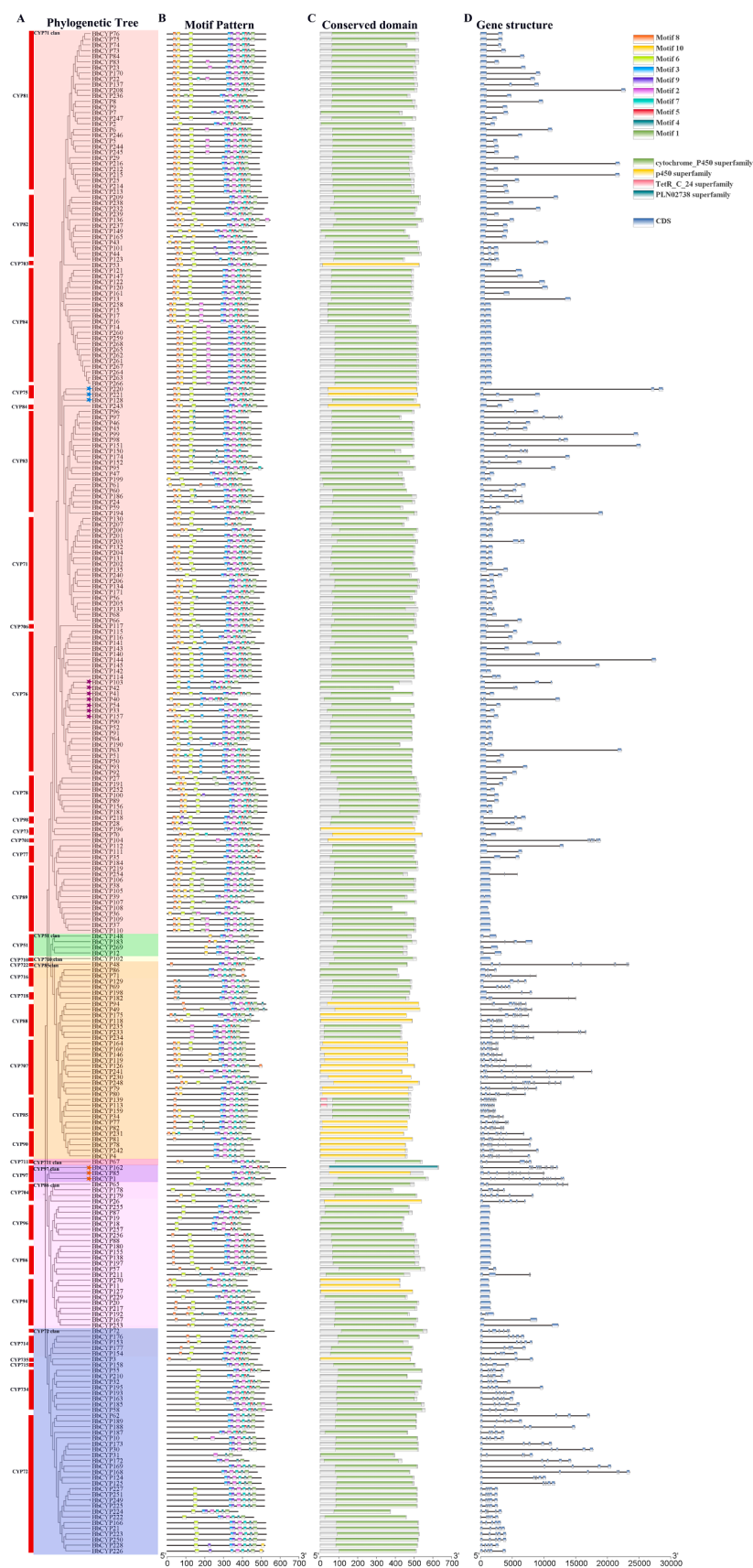


Fig. 1. Phylogenetic relationships of BbCYPs and CYPs from other plants. The genes in *Bougainvillea × buttiana* are marked with purple circle, while those in other plants are marked with green triangle. The known pigment formation-related CYPs are marked with red asterisk, while the putative pigment formation-related BbCYPs are marked in blue, purple and orange boxes. The different color arcs represent the different clans.



(caption on next page)

Fig. 2. Phylogenetic relationships, conserved motifs and domains, and gene structure of BbCYPs. (A) Phylogenetic analysis of BbCYP amino acid sequences using the neighbor-joining (NJ) method. The putative pigment formation-related BbCYPs are marked with asterisks. (B) Motif compositions of the BbCYP proteins. The different colored boxes indicate different motifs. The detailed information of each motif is shown in Fig. S1. (C) Conserved domains of the BbCYP proteins. The different colored boxes indicate different domains. (D) Gene structure of *BbCYP* genes. The blue box and black lines indicate the CDS regions and introns, respectively.

comprises 15 subfamilies, the CYP85 clan encompasses 7 subfamilies, the CYP72 clan with 5 subfamilies, and the CYP86 clan consists of 4 subfamilies.

3.3. Gene structure, motif composition and conserved domains of BbCYP proteins

To delve deeper into the structural attributes of BbCYPs, an analysis was conducted on their gene structure, motif composition, and conserved domains. The findings revealed that over half (50.37 %) of the *BbCYP* genes contained either 0 or 1 intron, with a further 47.78 % possessing between 2 and 9 introns, and a mere 1.85 % containing 10 or more introns. Intron number varied significantly across different families. For example, 64.37 % of the CYP71 clan members had one intron, 62.86 % of the CYP85 clan members had seven or eight introns, 76.92 % of the CYP72 clan members had four introns, and 65.38 % of the CYP86 clan members lacked introns. Notably, the CYP97 clan members exhibited a relatively high intron count (11–15), in contrast to the CYP86 clan members, which demonstrated a lower intron count (0–6). (Fig. 2D).

Over half of the BbCYPs (53.33 %) contain 10 motifs, while the remaining 46.67 % contain 3–9 motifs, as revealed by motif composition analysis (Fig. 2B). Distinct motif compositions characterize each clan, with CYP71 clan members harboring motifs 6–12 and 86.88 % containing ≥ 10 motifs. CYP72, CYP85, and CYP86 clan members contain motifs 3–9, 6–11, and 6–10, respectively. However proteins within the same subfamily exhibited similar motif composition, e.g., 89.65 % of CYP81 subfamily members contain 10 motifs, and 50.00 % of CYP72 subfamily members contain 7 motifs. Furthermore, utilizing the Batch CD-Search tool, we identified P450 or CypX conserved domains in 99.96 % of BbCYPs (Fig. 2C). Consequently, these BbCYPs are predicted to have P450 function, enabling their use for subsequent analysis. Additionally, BbCYPs within the same subfamily may share functional similarities.

3.4. Distribution on chromosomes and collinear analysis of BbCYP proteins

Previous reports have established that BBMB comprises 34 chromosomes (Lan et al., 2023). We have mapped 270 BbCYPs onto these chromosomes. Gene distribution occurs across all 34 chromosomes, albeit with varying numbers of BbCYPs on each. The chromosome 12

accommodates the highest gene number (24), while chromosomes 25 and 27 each contain only one BbCYP. Moreover, the distribution of *BbCYP* genes on the remaining chromosomes is as follows: chromosome 7 (19), chromosome 34 (16), chromosome 10 (15), chromosome 26 (14), chromosome 13 (13), chromosome 14 (12), chromosome 29 (11), chromosomes 2/15/16/21 (10 each), chromosome 1/11 (9 each), chromosome 20/28 (8 each), chromosome 9/22/32 (7 each), chromosomes 30/31/33 (6 each), chromosomes 3/6 (5 each), chromosomes 8/17 (4 each), chromosome 4 (3), and chromosomes 5/18/19/23/24 (2 each) (Fig. 3).

Gene collinearity analysis aims to identify gene sequence duplication events. Gene duplication studies hold significant implications for understanding species evolution and gene function diversity (Sun et al., 2020). Analysis of *BbCYP* genes duplication events revealed that 52 BbCYPs are tandem-duplicated genes, while 162 BbCYPs result from Whole Genome Duplication (WGD) or segmental duplication. A total of 417 collinearity pairs were identified among these BbCYPs (Fig. 4). Notably, segmental duplication events contribute substantially to the expansion of the BbCYP gene family. Moreover, collinearity among BbCYP family members extends beyond homologous chromosomes, involving many members of non-homologous chromosomes (Fig. 4). This broad distribution of BbCYP gene family members across chromosomes may account for this observation.

3.5. Collinear analysis of CYP proteins from *B. × buttiana* and other plants

To infer the evolutionary origin of the BbCYP family, a collinearity analysis was conducted between *B. × buttiana*, *G. max*, *A. thaliana*, *B. vulgaris*, *N. tabacum*, and *O. sativa* (Fig. 5). Numerous homologous gene pairs were identified. In *G. max*, *A. thaliana*, *B. vulgaris*, *N. tabacum*, and *O. sativa*, BbCYP genes had 170, 86, 124, 61, and 34 orthologous gene pairs, respectively. Notably, 12 BbCYP genes had orthologous gene pairs in all five plants. Among the 13 pigment synthesis-related BbCYP genes, 8 (*BbCYP1*, *BbCYP33*, *BbCYP54*, *BbCYP85*, *BbCYP128*, *BbCYP157*, *BbCYP162*, *BbCYP220*), 5 (*BbCYP85*, *BbCYP103*, *BbCYP128*, *BbCYP162*, *BbCYP220*), 2 (*BbCYP85*, *BbCYP220*), 2 (*BbCYP85*, *BbCYP220*), and 1 (*BbCYP128*) showed a syntenic relationship with those in *B. vulgaris*, *G. max*, *A. thaliana*, *N. tabacum*, and *O. sativa*, respectively. The results indicated a close relationship between *B. × buttiana* and *B. vulgaris*, and a more distant relationship between *B. × buttiana* and *O. sativa*.

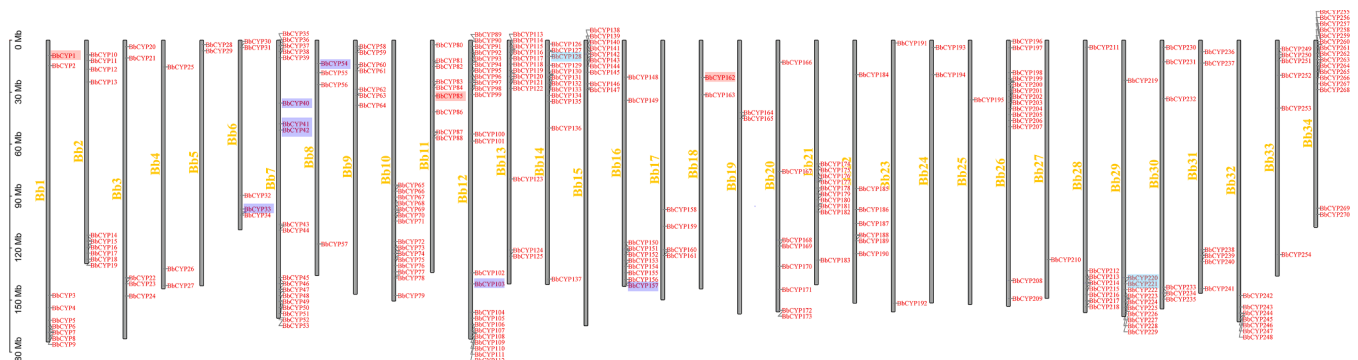


Fig. 3. Schematic representations of the chromosomal distribution of the BbCYP genes. The chromosome number is indicated to the left of each chromosome and gene number on the right. The putative pigment formation-related BbCYPs are marked in blue, purple and orange boxes.

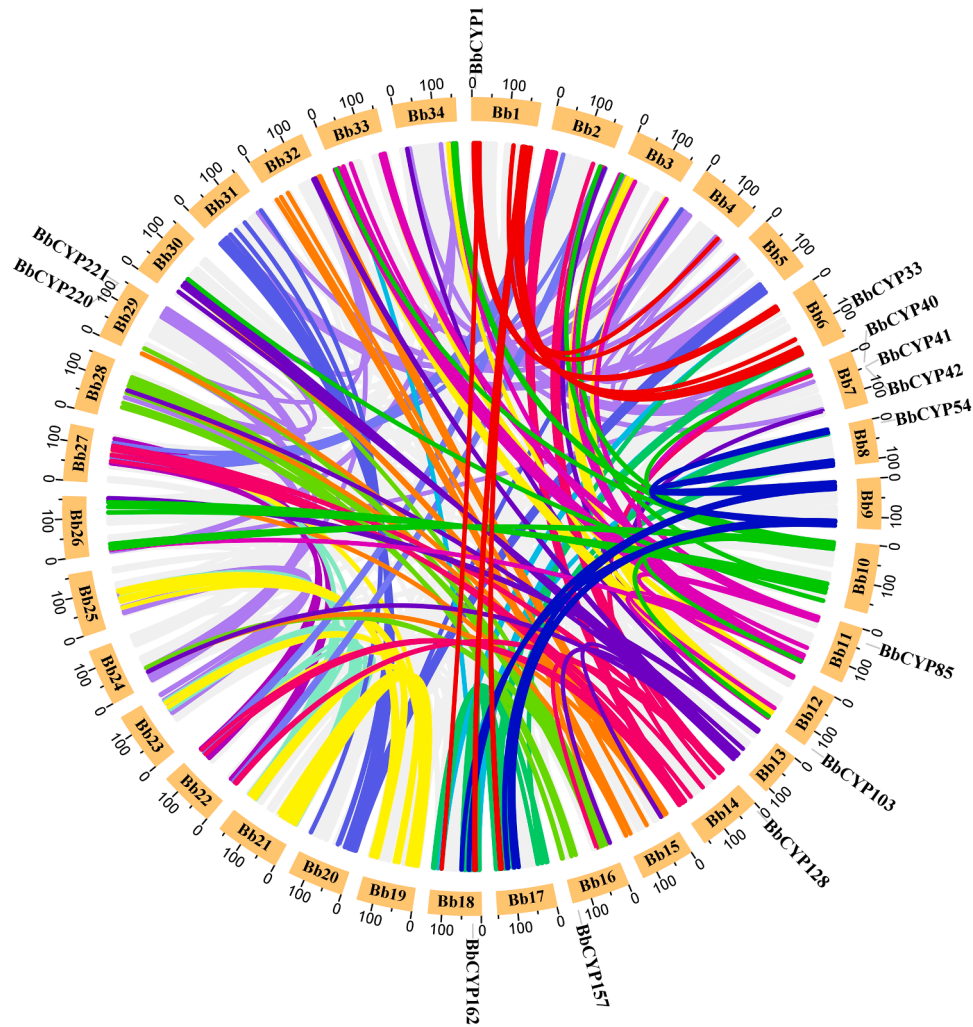


Fig. 4. Synteny analysis of *BbCYP* genes. The gray lines in the background represent all syntentic blocks while the colored lines represent syntentic blocks of *BbCYP*s in *B. x butiana* genome. The genes marked in the outermost layer of the circle are 13 pigment-related genes.

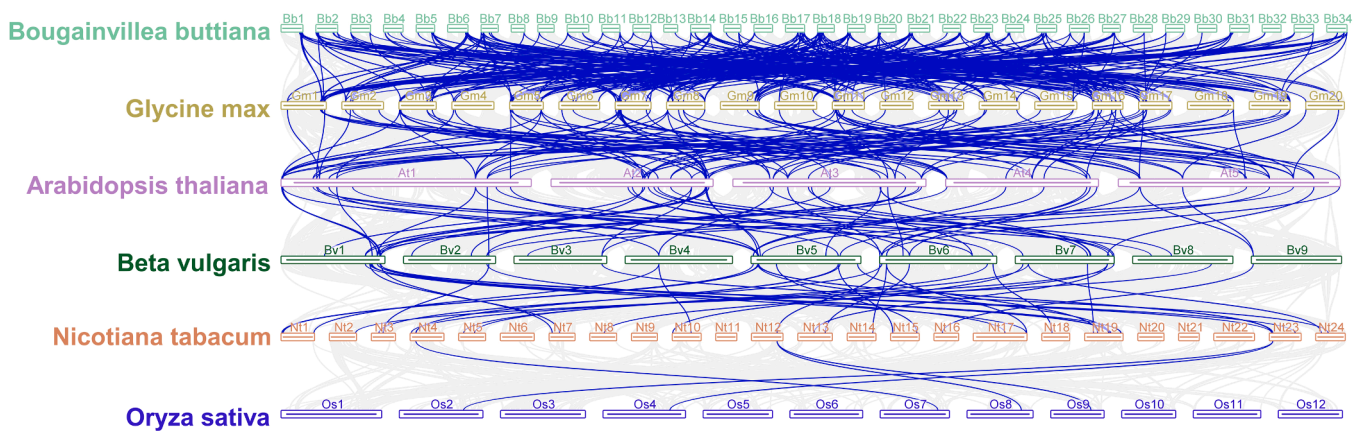


Fig. 5. Synteny analyzes between the CYP genes of *B. x butiana* and five representative plant species. The gray lines in the background indicate collinear blocks within the genomes of *B. x butiana* and other plant species, while the blue lines highlight syntenic CYP gene pairs.

3.6. Expression patterns of 13 *BbCYP* genes in bracts at three developmental stages

To validate the roles of *BbCYP* genes in pigment synthesis, we analyzed their expression levels in BBMB bracts across three

developmental stages using real-time PCR. The expression of CYP75 clan and CYP76AD clan genes progressively increased. *BbCYP1*, *BbCYP162*, and *BbCYP85*, as homologous genes of CYP97 clan, exhibited distinct patterns. *BbCYP1* was not expressed, *BbCYP162* exhibited relatively low expression levels, while *BbCYP85* showed an upward trend (Fig. 6). The

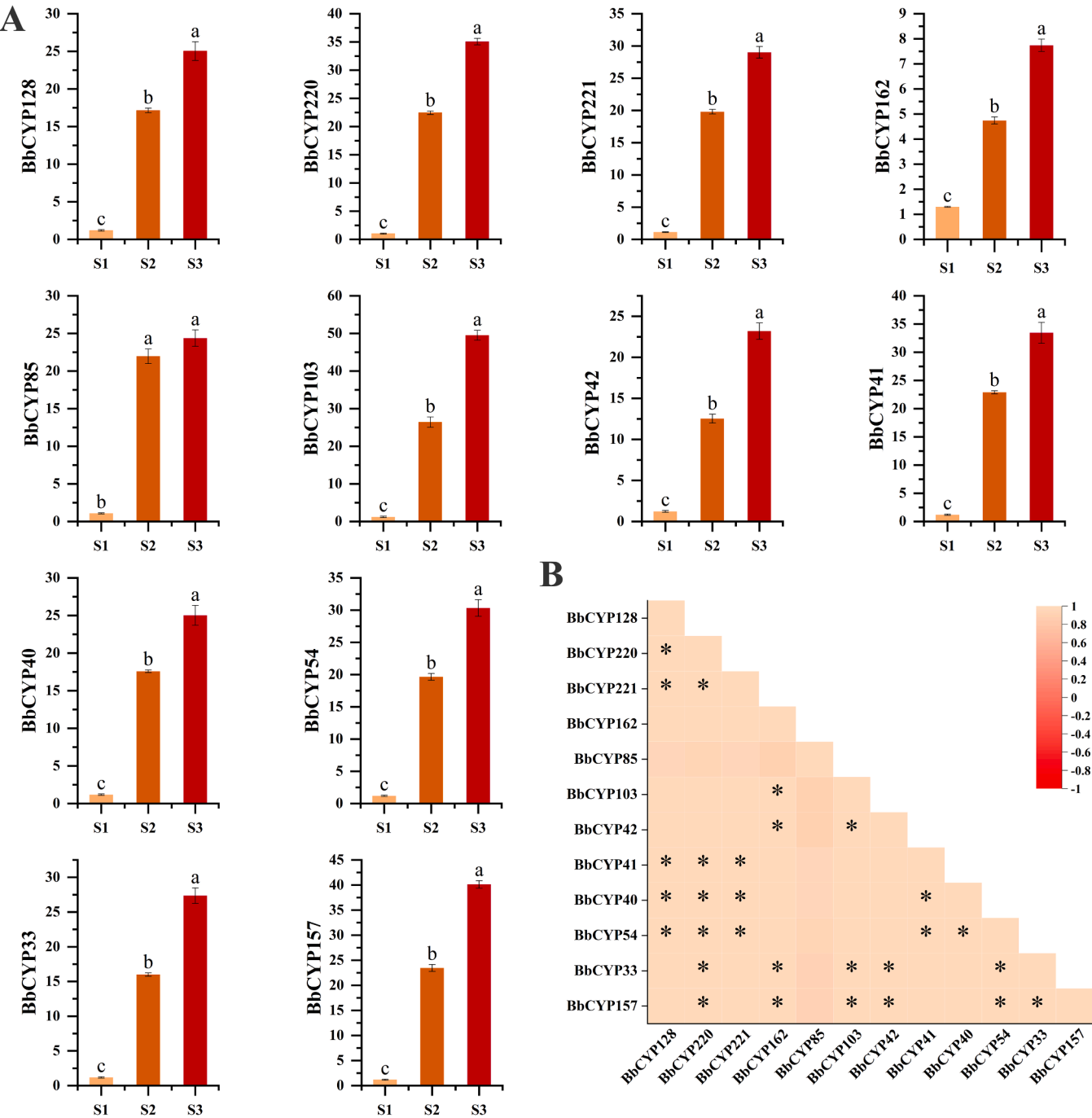


Fig. 6. Expression analysis of the 13 *BbCYP* genes in *B. × buttiana* at different development periods. (A) The qPCR assay was used to examine the expression patterns of 12 *BbCYP* genes in bracts at three developmental periods. Error bars were obtained from three measurements. Small letter(s) above the bars indicate significant differences ($P < 0.05$) among the samples. (B) The correlation between the gene expression of *BbCYP*s. Light brown: positively correlated; Red: negatively correlated. The black asterisk indicates a significant correlation at the 0.05 level.

correlation analysis showed that most genes display significant positive correlation, except *BbCYP85*. These findings suggest that betalains and flavonoids may contribute more significantly than carotenoids to the color formation of *B. × buttiana* bracts.

3.7. Pigment contents and color parameters changes of bracts at three developmental stages

Furthermore, we detected changes in pigment contents and color parameters of bracts at three developmental stages. Fig. 7A showed that flavonoids and betalains content increased, while carotenoids content

initially increased and then decreased. The L, a, and C* values exhibited an upward trend, and b and H* values a downward trend (Table S3). This suggests that as BBMB grows, its bracts become redder and brighter. To screen genes related to pigment synthesis, we analyzed the correlation between pigment contents and gene expression levels, along with the correlation between pigment contents and color parameters. The expression level of *BbCYP40* exhibited the strongest correlation with betalains content, indicating that it may play an important role in betalain biosynthesis. Similarly, *BbCYP220* may play pivotal role in flavonoids biosynthesis. Besides, *BbCYP85* may play a relatively important role in carotenoids biosynthesis (Fig. 7B). From Fig. 7C, we

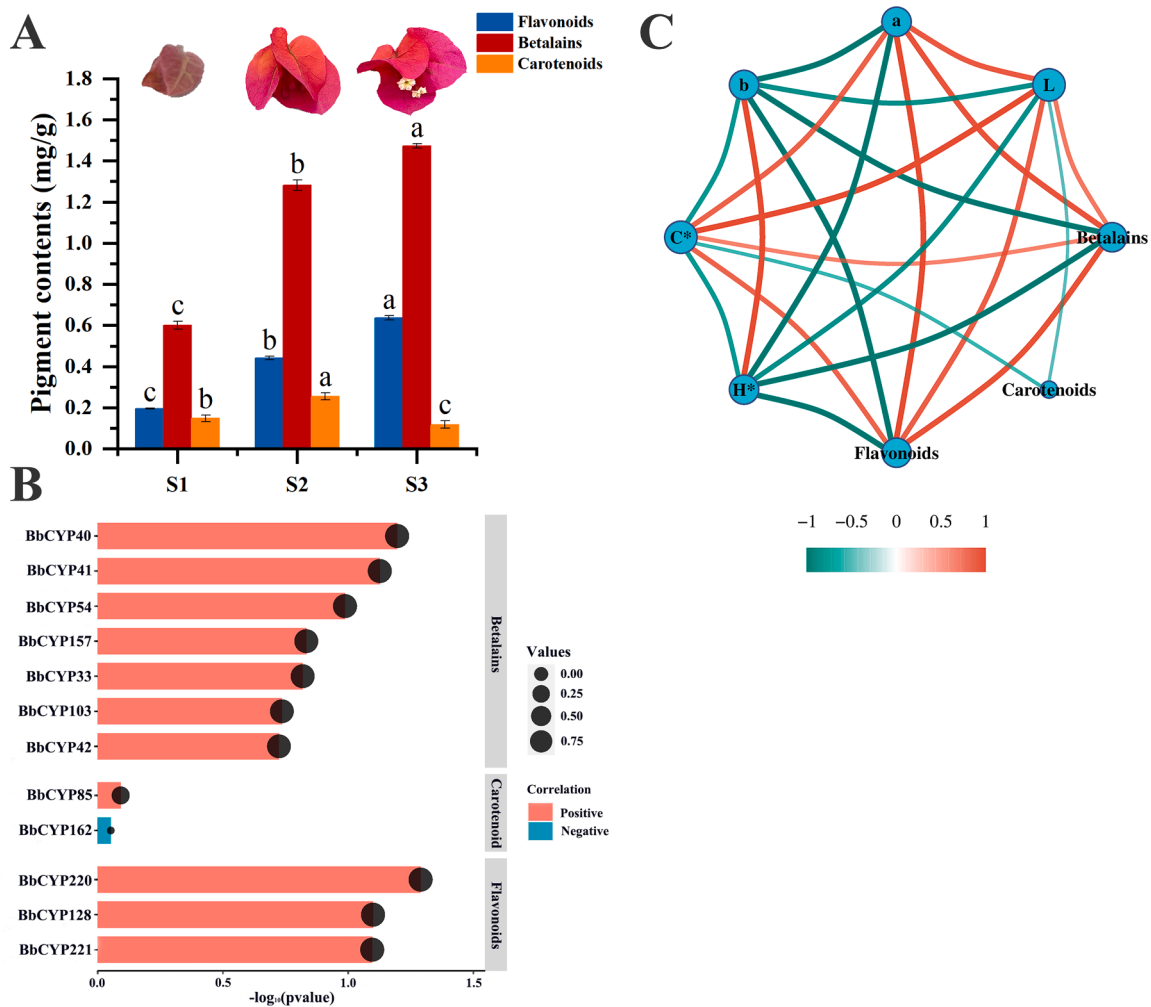


Fig. 7. Analysis of flavonoids, betalains, and carotenoids content in bracts at three developmental periods and correlation analysis. (A) The pigments contents in bracts at three developmental periods. Small letter(s) above the bars indicate significant differences ($P < 0.05$) among the samples. (B) The correlations between gene expression levels and pigments contents in bracts. Red: positively correlated; Green: negatively correlated. (C) The correlations between pigments contents and color parameter changes in bracts. Red: positively correlated; Green: negatively correlated.

can see that flavonoids and betalains content positively correlated with L, a, and C* values, but negatively with b and H* values. Carotenoids content only negatively correlated with L and C* values. Overall, flavonoids and betalains are more important than carotenoids in *Bougainvillea* bracts color formation. *BbCYP40* and *BbCYP220* could potentially be utilized in the breeding of novel-colored *Bougainvillea*.

4. Discussion

4.1. Identifying *BbCYP* gene families at the genome-wide level

The availability of reference genomes has enabled the comprehensive identification and analysis of gene families at the genome-wide level, providing valuable genetic resources for improving plant characteristics. This method has identified numerous plant gene families, such as the genome-wide identification of *Tartary buckwheat* CYP genes by Sun et al. (2020), which preliminarily identified potential CYP genes regulating fruit size. Park et al. (2023) performed a genome-wide search for the AP2/ERF family in *Lactuca sativa* and assessed the expression of the AP2/ERF genes under various stress conditions. The results provide valuable potential molecular targets for improving stress responses in lettuce. Similarly, Sharma et al. (2020) identified the U-box E₃ ubiquitin ligase gene family in tomato, suggesting that the U-box gene family is involved in tomato plant development and stress responses. In this

study, to investigate the potential functions of P450 superfamily proteins in the pigment synthesis of *Bougainvillea*, we investigated the P450 superfamily genome wide. Our analysis offers a comprehensive understanding of the evolutionary relationships, chromosomal distribution, structural features, and duplication events of the CYP gene family in *Bougainvillea* (Fig. 2, Fig. 3 and Fig. 4). These results provide valuable insights into the evolutionary mechanisms of the CYP gene family in *Bougainvillea*.

4.2. Patterns of characteristics evolved in the CYP genes of *Bougainvillea*

Cytochrome P450 s (P450 s) are a family of hemoproteins that catalyze various reactions in plant growth and development (Montellano, 2005). They have been identified in many plant species, but the number and structural features of P450 s vary. For instance, there are 272 CYP genes in *A. thaliana* (divided into nine clans) (Paquette et al., 2000), 276 in *Osmanthus fragrans* (divided into ten clans) (Liu et al., 2022), 285 in *Tartary buckwheat* (divided into eight clans) (Sun et al., 2020), 299–419 in Chinese white pear (divided into 10 clans) (Zhang et al., 2023), and 174 in *Morus notabilis* (divided into nine clans) (Ma et al., 2014). In this study, we identified 270 CYP genes in *Bougainvillea*, divided into eight clans. Most plant species contain 100–300 CYPs, and the number of family members is independent of genome size. The CYP71 clan has the largest number of genes (160

BbCYP genes) and is mainly involved in core metabolisms, essential metabolisms, and specialized compound biosynthesis. No *BbCYP* genes belong to the CYP74 clan, possibly due to gene loss during evolution (Fig. 1).

The P450 family reportedly contains four highly conserved motifs: heme-binding domain, I-helix region, K-helix domain, and PERF motif (Li et al., 2007). Analysis of *BbCYPs* using MEME revealed the presence of these motifs, with motif 1 representing the heme-binding domain, motif 2 representing the K-helix domain, motif 3 representing the I-helix region, and motif 4 representing the PERF motif (Fig. S1). All *BbCYPs* contain these motifs, and the same clan exhibited similar motif components, suggesting that genes within the same family have similar functions. This allows for the screening of candidate genes based on phylogenetic classification.

The results of gene structure analysis indicated that half of *BbCYP* genes have a simple gene structure, with only one (10.7 %) or two exons (39.6 %) (Fig. 2). Previously, Stephens et al. (2020) reported that single-exon genes enhance total expression dosage in stress responses to ensure plant survival in extreme environments. As *Bougainvillea* is considered drought-tolerant, salt-tolerant, and pollutant-tolerant (Salam et al., 2017; Sharma et al., 2005), the simple structure of *BbCYP* genes may be involved in *Bougainvillea* stress responses.

4.3. *BbCYP40* and *BbCYP220* may play important roles in *Bougainvillea* bract color formation

The aim of this study was to identify CYP genes involved in pigment biosynthesis pathway at genome wide level and lay a foundation for future cultivation of novel-colored *Bougainvillea*. Tanaka and Brugliera (2013) reported the modification of flower color using Cytochromes P450 via metabolic engineering. And previous reports have identified the involvement of CYP75B (flavonoid 3'-hydroxylase, F3'H), CYP75A (flavonoid 3',5'-hydroxylase, F3',5'H), CYP93B (flavone synthase II, FNSII), CYP97A/C (β -hydroxylases), and CY76AD in the formation of anthocyanins, carotenoids, and betalains, respectively (Tanaka and Brugliera, 2014). Given that genes within the same family or with similar structures tend to exhibit the similar function, we selected 13 *BbCYP* genes potentially related to *Bougainvillea* pigment synthesis, based on evolutionary relationships and gene structure analysis (Fig. 1 and Fig. 2). These included three genes corresponding to F3'H, three genes corresponding to β -hydroxylases, and seven genes corresponding to CYP76AD, with no genes corresponding to F3',5'H and FNS.

Previous studies showed that the relative proportion of the pigments determines the variation in bract colour of *Bougainvillea* (Salam et al., 2017). The association analysis between pigments contents and changes in color parameters during bract development indicated that betalains and flavonoids may play major roles in *Bougainvillea* $\times$ *buttiana* 'Mrs Butt' bracts color formation (Fig. 7C). *BbCYP40*, as homologous genes of CYP76AD, exhibited the strongest correlation with betalains. As a homologous gene of F3'H, *BbCYP220* display the strongest correlation with flavonoids (Fig. 7B). The results speculated that *BbCYP40* and *BbCYP220* play significant roles in the formation of bract colors in *Bougainvillea*.

In previous studies, we identified DOD (Cao et al., 2018), cDOPA 5GT (Liu et al., 2021), CHS (Sun et al., 2021), DFR (Sun et al., 2024), and F3H (Sun et al., 2022) genes in *Bougainvillea*. The results of this study provides further opportunities for the cultivation of novel-colored *Bougainvillea*.

5. Conclusions

The present study conducted a genome-wide identification and analysis of the CYP450 family genes in *Bougainvillea*, with a special focus on *BbCYP* genes involved in pigmentation. We identified 13 potential *BbCYPs* homologous to F3'H, CYP97, and CYP76AD. The correlation analysis results indicated that among the 13 candidate *BbCYPs*,

BbCYP40 and *BbCYP220* may play major roles in *Bougainvillea* bract color formation. Overall, this study provides valuable reference materials for the cultivation of novel *Bougainvillea* bract colors.

CRedit authorship contribution statement

Rong Sun: Writing – original draft, Validation, Investigation, Funding acquisition, Conceptualization. **Shan Liu:** Writing – review & editing, Visualization, Investigation. **Yi Zheng:** Writing – review & editing, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no competing interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scienta.2025.113997.

Data availability

The authors do not have permission to share data.

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